

Programme: B E – Department of Artificial Intelligence and Data Science
Internal Assessment – I (Re-Test)

TERM : 15/04/24 to 27/07/24	COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS
DATE : 13-07-24 TIME: 12:00 to 1:00	COURSE CODE : AD43
MAX MARKS: 28	PORTIONS : L1-L18

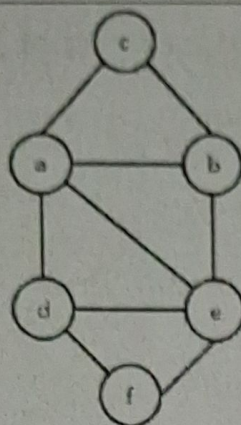
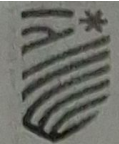


Mobile Phones are banned

Instructions to Candidates: *Answer any 2 questions.*
~~Q1 compulsory and Choice between Q2 and Q3.~~

Marks 14x2=28

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks																																								
1.a	Identify the stable matching set S for the following set of men and women using Gale-Shapley algorithm. Men's preference list Women's preference list <table border="1"> <tr><td>W</td><td>H</td><td>I</td><td>J</td><td>K</td></tr> <tr><td>X</td><td>I</td><td>J</td><td>H</td><td>K</td></tr> <tr><td>Y</td><td>J</td><td>H</td><td>I</td><td>K</td></tr> <tr><td>Z</td><td>H</td><td>I</td><td>J</td><td>K</td></tr> </table> <table border="1"> <tr><td>H</td><td>X</td><td>Y</td><td>Z</td><td>W</td></tr> <tr><td>I</td><td>Y</td><td>Z</td><td>W</td><td>X</td></tr> <tr><td>J</td><td>Z</td><td>W</td><td>X</td><td>Y</td></tr> <tr><td>K</td><td>W</td><td>X</td><td>Y</td><td>Z</td></tr> </table>	W	H	I	J	K	X	I	J	H	K	Y	J	H	I	K	Z	H	I	J	K	H	X	Y	Z	W	I	Y	Z	W	X	J	Z	W	X	Y	K	W	X	Y	Z	L3	CO1	05
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b	i. Construct a proof for "If $f(n) = \theta(g)$ and $g(n) = \theta(h)$ then $f(n) = \theta(h)$" with an example. ii. Construct a proof for "If f is a polynomial of degree d, in which the coefficient a_d is positive. Then $f = O(n^d)$."	L3	CO2	05																																								
c	Describe the notion of instability w.r.t to Stable matching with an example.	L2	CO1	04																																								
2.a	Consider two players P1 and P2 and nodes in the following order for a competitive facility location problem. [10,2,3,10,5,20,25,10,5]. If P1 moves first and then P2, estimate whether target bound 10 and 20 are achievable by P2??	L3	CO2	05																																								
b	Describe an algorithm for graph colouring using BFS approach.	L2	CO2	05																																								
c	What do you mean by efficiency of an algorithm? Explain the approach to determine the efficiency with an example.	L2	CO1	04																																								
3.a	Identify the DFS traversal for the following graph.	L3	CO2	05																																								



b	Examine the time complexity of the following: <pre>int fun(int n) { int count = 0; for (int i = 0; i < n; i++) for (int j = i; j > 0; j--) count = count + 1; return count; }</pre> ii. <code>void fun(int n, int arr[])</code> <pre>{ int i = 0, j = 0; for(; i < n; ++i) while(j < n && arr[i] < arr[j]) j++; }</pre>	L3	CO1	04
c	Describe a greedy algorithm to schedule the given intervals with each interval having a start time s_i and finish times f_i .	L3	CO3	05